

Paper Information Record — Hu2025

Paper ID: 0021 | Oxford Missing Data Study

General Information

Field	Value
Paper ID	0021
Title	Corporate Boards and Innovation: Director Expertise and Firm-Level R&D
Authors	Hu, S., et al.

Data Structure

Field	Value
Data type	Firm-year panel, two-way FE (gvkey + year), clustered SEs (gvkey)
Observations (N)	21,882 (regression sample after FE absorption)
Preferred model column	Main DiD specification

Core Regression Model

Field	Value
Dependent variable	n1pfdir1 (director independence measure)
Focal IV	post1_x_treat1 (DiD interaction: post × treatment)
Controls	See simulation_0021.py Config.CONTROLS
Fixed effects	gvkey + year (two-way)
Clustered SEs	gvkey (firm-level)

Simulation Configuration

Field	Value
Key variables (4)	ptita, mb, lev, aret
MAR control	tio1 (institutional ownership, continuous, complete)
Mechanisms	MCAR, MAR, NMAR
Missingness proportions	1%, 5%, 10%, 20%, 30%, 40%, 50%
Methods	LD, Mean, Reg, Iter, RF, DL, MILGBM
Iterations per scenario	30
MAR/NMAR strength	1.5
M (multiple imputation)	5

Baseline Validation

Field	Value
Baseline coefficient	0.0331 (post1_x_treat1)
Baseline SE	0.00457

Field	Value
Baseline p-value	5.78e-13
Published value	0.033*** (highly significant DiD)
R-squared	See simulation output

Run Log

Field	Value
Full run start	2026-04-11
Full run complete	2026-04-13
Total combinations	17640 (3x7x4x30x7)
Errors	0
DL method	TensorFlow 2.21.0, MLP 32->Dropout(0.1)->16->1, ReLU, Adam lr=0.005, EarlyStopping
MILGBM method	LightGBM 4.6.0, MICE M=5, Rubin's Rules, 30 estimators, max_depth=4
Note	Run interrupted and resumed; summary sheets rebuilt from parsed txt outputs (rebuild_summary=True).